

Name: _____

Unearthing Mars — A Historical Perspective

Student Answer Packet 1: To be completed with a partner. Numbers next to each task indicate the steps found in your **Student Directions** handout. Express all numerical answers to the nearest tenth unless indicated.

2. Eccentricity of *Model of Mars' Orbit Using Given Calculations*

$\text{Eccentricity} = \frac{\text{distance between foci (cm)}}{\text{length of major axis (cm)}}$
Solve for x : $0.094 = \frac{x}{16.1 \text{ cm}}$
Workspace:
Distance between foci: _____ cm

13. Length of major axis: _____ cm

14. a. M1 to Sun: _____ cm

b. Sun to M3: _____ cm

d. Distance between foci: _____ cm

15. Eccentricity of *Model of Mars' Orbit Using Triangulation*

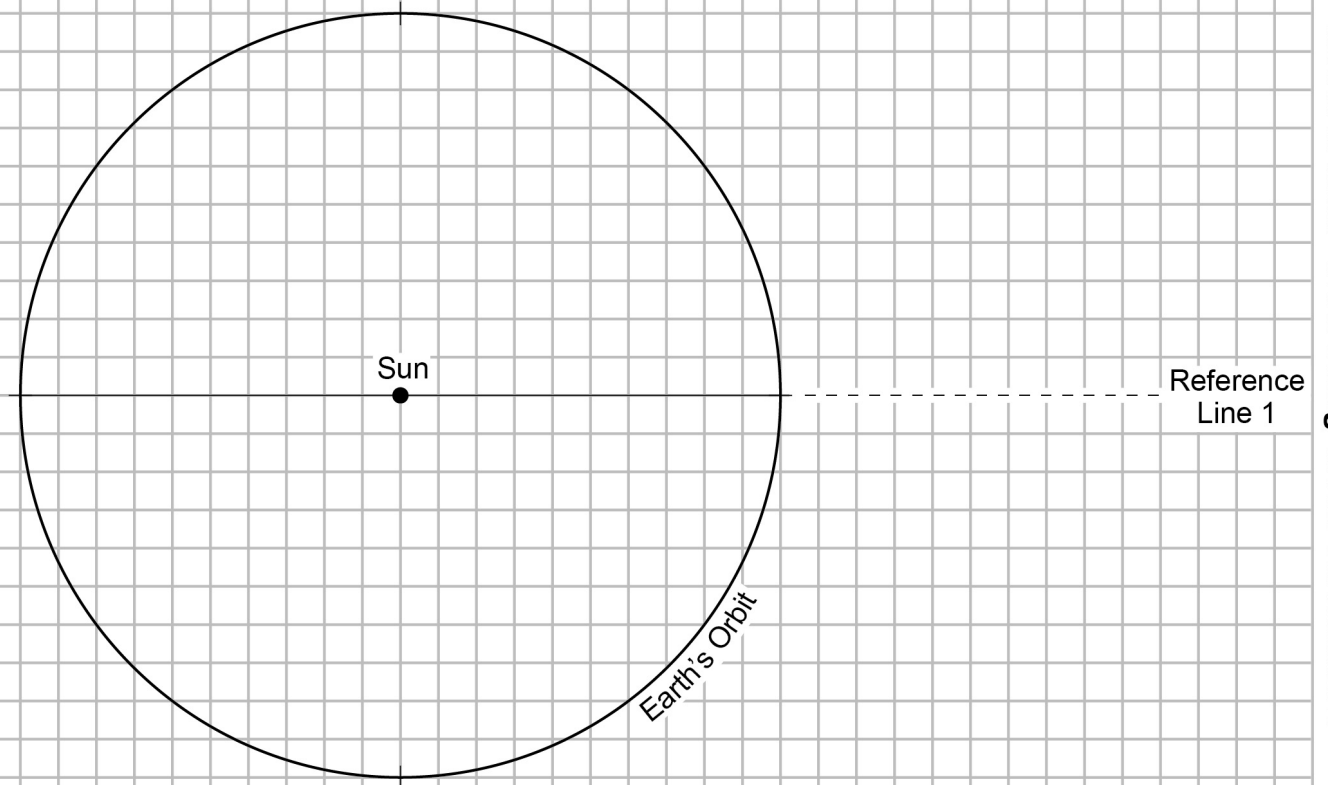
$\text{Eccentricity} = \frac{\text{distance between foci (cm)}}{\text{length of major axis (cm)}}$
Workspace:
Eccentricity: _____ (express your answer <i>to the nearest thousandth</i>)

16. Percent error calculation

$\% \text{ Error} = \frac{\text{calculated value} - \text{accepted value}}{\text{accepted value}} \times 100$
Workspace:
% Error: _____ %

Name _____

Model of Mars' Orbit Using Given Calculations



Name _____

Model of Mars' Orbit Using Triangulation

